

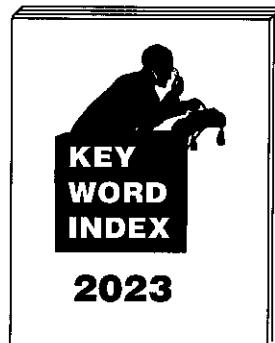
2023

JOURNEYMAN

OPEN

BOOK

FINAL EXAM #4



70 QUESTIONS

TIME LIMIT - 3 HOURS

TIME SPENT **MINUTES**

SCORE **%**

Journeyman Final Exam #4 70 Questions - Time Limit 3 Hours

1. When applying the demand factors of Table 220.56, in no case can the feeder or service demand load be less than the sum of the ____.

- (a) largest two kitchen equipment loads (b) heat and air conditioning loads
(c) rating of the appliance loads (d) total receptacle load

2. The ampacity of capacitor circuit conductors must not be less than ____ of capacitor current.

- (a) 80% (b) 100% (c) 125% (d) 135%

3. The number of conductors allowed in LFNC must not exceed that permitted in ____.

- (a) Table 7 (b) Appendix D (c) Table 1 (d) Table 4

4. A single piece of equipment consisting of a multiple receptacle comprised of ____ or more receptacles must be computed at not less than 90 va per receptacle.

- (a) one (b) two (c) three (d) four

5. Per the Code, the volume of a 3" x 2" x 2" device box is ____ cubic inches.

- (a) 10 (b) 12 (c) 14 (d) 15

6. The AC ohms-to-neutral impedance per 1,000 feet of #4/0 aluminum in a steel raceway is ____.

- (a) 0.05Ω (b) 0.010Ω (c) 0.101Ω (d) 0.10Ω

7. A value assigned to a circuit or system for the purpose of conveniently designating its voltage class such as 120/240 volt is called ____ voltage.

- (a) average (b) source (c) nominal (d) effective

8. A 240 volt single-phase room air conditioner shall be considered as a single-phase motor unit if its rating is not more than ____.

- (a) 20 amps (b) 25 amps (c) 30 amps (d) 40 amps

9. A steel cable tray of .79 square inches is used as an equipment ground conductor. The maximum rating of the circuit breaker permitted for this application is ____ amps.

- (a) 1000 (b) 600 (c) 200 (d) 400

10. A bathroom in a dwelling has a counter space of seven feet including the sink. How many receptacles are required to serve this area?

- (a) 1 (b) 3 (c) 4 (d) none are required**

11. An installation requires a device box with a capacity of 10.25 cubic inches. What is the minimum size box allowed?

- (a) 2" x 2" x 3" (b) 3" x 2" x 2 1/4" (c) 3" x 2" x 2" (d) 2" x 3" x 3"**

12. A show window is calculated at ____ va per linear foot.

- (a) 180 (b) 1500 (c) 1800 (d) 200**

13. The maximum number of 15 amp receptacles permitted on a free standing office partition is ____.

- (a) 10 (b) 13 (c) 2 (d) 6**

14. Several motors, each not exceeding 1 horsepower in rating, shall be permitted on a nominal 120 volt branch circuit protected at not over ____ amperes.

- (a) 15 (b) 20 (c) 30 (d) 40**

15. At least one receptacle outlet per unit, accessible at grade level, shall be installed at the front and back on ____.

- (a) commercial buildings (b) office buildings only**
(c) motels (d) one and two family dwellings

16. Which of the following receptacles may be connected to the small appliance branch circuit?

- (a) Outside receptacles**
(b) Electric clock in a dining room
(c) Hallway receptacle which is within 20 feet of kitchen
(d) Garage ceiling receptacle for automatic garage door opener

17. Where the ampacity of a conductor does not correspond with the standard ampere rating of a fuse or circuit breaker, the next higher standard size is permitted only if the rating does not exceed ____ amperes.

- (a) 600 (b) 800 (c) 1,000 (d) 1,200**

18. Where wet contact is likely to occur in a permanent amusement attraction, ungrounded 2-wire DC control circuits shall be limited to ____ maximum for continuous DC or 12.4 volts peak for DC that is interrupted at a rate of 10 to 200 Hz.

- (a) 12 volts (b) .031 va (c) 24 watts (d) 30 volts

19. It shall be permissible to apply a demand factor of 75% to the nameplate-rating load of 4 or more ____ fastened in place in a dwelling.

I. water heaters II. dishwashers III. clothes dryers

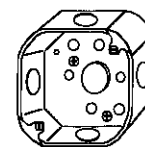
- (a) I only (b) II only (c) I and II only (d) I, II and III

20. When more than one calculated or tabulated ampacity could apply for a given circuit length, the ____ value shall be used.

- (a) lowest (b) average (c) highest (d) none of these

21. What size octagon box is required for 5 - #12 and 4 - #14 conductors?

- (a) 1 1/4" (b) 1 1/2" (c) 2" (d) 2 1/8"



22. Assuming that the area of all the conductors (over 4 conductors) to be installed in a 1" EMT conduit is 0.30 sq. in., which of the following is true?

- (a) The conductors' area is greater than the allowable area of tubing fill.
(b) The conductors' area is less than the allowable area of tubing fill.
(c) It is necessary to select another trade size.
(d) It is necessary to change the THW conductors to RHW.

23. When sizing the service conductors for an apartment complex, the minimum demand load in kW for eight 4 kW ranges would be ____ kW.

- (a) 11.52 (b) 16.96 (c) 32 (d) None of these

24. The ampacity of a #14 THW conductor, when there are six conductors in a conduit and the temperature is 30° C, would be ____ amps.

- (a) 25 (b) 22 (c) 20 (d) 16

25. In a custom house, the demand load (load applied for service calculation) for a 12 kW range and a 4 kW oven is ____ kW.

- (a) 11 (b) 11.2 (c) 12 (d) 16

26. The branch circuit conductor supplying a 3/4 hp, 1 ϕ 115 volt motor shall have an ampacity of at least ____.
- (a) 13.8 amps (b) 17.25 amps (c) 20 amps (d) 21.3 amps
27. You are wiring a house that has 2200 square feet under the roof. The living area accounts for 2000 square feet of this space. The minimum general lighting load for this dwelling would be ____ VA.
- (a) 6000 (b) 6600 (c) 7700 (d) 8000
28. When sizing the service on a dwelling unit, the small appliance load plus laundry load should be computed at ____ VA.
- (a) 1,500 (b) 3,000 (c) 4,500 (d) 6,000
29. #2/0 THW copper service conductors would require a grounding electrode conductor of ____.
- (a) #6 (b) #4 (c) #3 (d) #2
30. The feeder carrying two single phase, 1 1/2 hp, 230v motors would be required to have a load current rating of ____ amps.
- (a) 12 (b) 16 (c) 22.5 (d) 30
31. What is the allowable ampacity of a #12 TW copper conductor in a raceway with an ambient temperature of 75°F?
- (a) 20 amps (b) 25 amps (c) 21.6 amps (d) 30 amps
32. The branch-circuit protection for a 1 ϕ , 115v, 3 hp motor would normally be ____ using dual element time delay fuses.
- (a) 35 amps (b) 40 amps (c) 60 amps (d) 90 amps
33. The ampacity of a #14 THW conductor, when there are six conductors in a conduit and the temperature is 30°C, would be ____ amps.
- (a) 25 (b) 22 (c) 20 (d) 16
34. The maximum motor-running overload protection would be ____ amps for a 3 hp 1 ϕ 230V motor.
- (a) 15 (b) 20 (c) 22.1 (d) 29.1

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35. The correction factor (C.F.) for 104°F is ____ for a 60°C insulated conductor.

- (a) 0.80 (b) 0.82 (c) 0.91 (d) 0.90

Half way through the exam

36. The branch-circuit load for one 6 kW oven would be ____ kW.

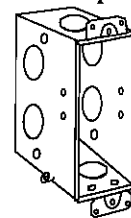
- (a) 6 (b) 4.8 (c) 4.2 (d) 5

37. If the full load current of a 2 hp, 115v, 1 ϕ motor with a service factor of 1.2 is 24 amps. What is the maximum motor running overload relay protection?

- (a) 24 (b) 27.6 (c) 31.2 (d) 33.6

38. What is the cubic inch capacity required for a device box containing one duplex receptacle, cable clamps and two #12-2 with ground nonmetallic sheathed cables (romex)?

- (a) 13.5 cu.in. (b) 15.75 cu.in. (c) 16 cu.in. (d) 18 cu.in.



39. The branch circuit conductor supplying a 10 hp 3 ϕ 230v motor shall have an ampacity of at least ____ amps.

- (a) 38.5 (b) 40.25 (c) 35 (d) 17.5

40. There are seven current carrying #12 THHN conductors run in a conduit where the ambient temperature is 30°C. What is the ampacity of each conductor?

- (a) 21 (b) 17.5 (c) 16 (d) 14

41. The ampacity of a #12 THW conductor when there are not more than three conductors in a raceway and the ambient temperature is 28°C, would be how many amps?

- (a) 20 (b) 25 (c) 30 (d) 35

42. A 10 unit apartment building has a 4.5 kW water heater in each unit. What is the demand on the service for these water heaters? Use standard method of calculation.

- (a) 45 kW (b) 50 kW (c) 33.75 kW (d) 36 kW

43. The overcurrent protection for a #10 RHH conductor in a conduit at 104°F is ____ amps.

- (a) 30 (b) 36.4 (c) 40 (d) 20

44. Under the optional calculation, the air conditioning is added to the service at ____.

- (a) 125% (b) full load (c) 80% (d) demand load**

45. A feeder supplying two 4 kW wall-mounted ovens and a 5 kW counter-mounted cooking unit shall have a demand of ____ kW.

- (a) 8.4 (b) 7.15 (c) 13 (d) none of these**

46. What is the feeder neutral demand for six - 4.5 kW household clothes dryers?

- (a) 18.9 kW (b) 21 kW (c) 27 kW (d) 15.7 kW**

47. A dwelling has a total of 2500 sq.ft., how many 15 amp circuits are required for the general lighting?

- (a) 2 (b) 3 (c) 4 (d) 5**

48. An owner wishes to add 12 receptacle outlets to the drawings of his residence. What load will these additional receptacles add to the service load?

- (a) 36va (b) 1200va (c) 2160va (d) none of these**

49. The demand load in kW for 15 - 3.5 kW ovens in an apartment complex would be ____ kW.

- (a) 9.6 (b) 16.8 (c) 30 (d) none of these**

50. The approximate area in square inches of a #14 RHH conductor without outer cover is ____ square inches.

- (a) 0.0209 (b) 0.0293 (c) 0.026 (d) 0.0097**

51. The branch circuit demand for one 8 kW range in a residence would be ____ kW.

- (a) 12 (b) 8 (c) 6.4 (d) 5.2**

52. What is the general lighting load for a single dwelling that measures 1400 sq.ft. inside and has an outside dimension of 1800 sq.ft.?

- (a) 4200va (b) 4500va (c) 4800va (d) 5400va**

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53. What size dual-element fuse does the Code require for a 2 hp, 208 volt, single-phase motor?
(a) 20 amp (b) 30 amp (c) 35 amp (d) 40 amp
54. The volume required for two #12 TW grounding conductors and two #12 TW conductors in a box would be ____ cubic inches.
(a) 9 (b) 6.75 (c) 6 (d) 4.5
55. What is the minimum size copper equipment grounding conductor required for a 80 amp circuit breaker?
(a) #12 (b) #10 (c) #8 (d) #6
56. What is the maximum size overload relay permitted for protection against overload for a 1 ϕ , 2 hp, 208 volt motor?
(a) 15.18 (b) 16.5 (c) 17.16 (d) none of these
57. The total area of 6 - #8 XHHW conductors and 2 - #6 XHHW conductors is ____ square inches.
(a) 0.3026 (b) 0.3265 (c) 0.3295 (d) 0.3802
58. The maximum current on the neutral with either Line one or Line two on is ____ amps if the load on Line one to neutral is 9000 watts and the load Line two to neutral is 9500 watts. The voltage is 120/240v.
(a) 75 (b) 77 (c) 79 (d) 154
59. How much space remains to be filled without exceeding the percentage allowed when nine #12 XHHW conductors are placed in a 1" rigid metal conduit?
(a) 0.1297 sq.in. (b) 0.1831 sq.in. (c) 0.1921 sq.in. (d) 0.0181 sq.in.
60. How many #8 TW conductors can be installed in a 1" IMC conduit, 18" long?
(a) 6 (b) 8 (c) 11 (d) 13
61. A 5 hp, single-phase, 230 volt, wound rotor motor would have nontime delay fuses sized at ____ percent.
(a) 300 (b) 175 (c) 150 (d) 250

62. How many #6 XHHW can be installed in a 2" PVC schedule 80 conduit?

- (a) 16 (b) 18 (c) 19 (d) 20

63. What is the ampacity of each conductor of a group of nine #14 RHH all in one conduit with an ambient temperature of 45°C?

- (a) 25a (b) 15.225a (c) 17.5a (d) 20a

64. What is the minimum volume required for a box containing three - #12 THW conductors and three - #10 THW conductors?

- (a) 10.5 cu.in. (b) 12.75 cu.in. (c) 14.25 cu.in. (d) 14.5 cu.in.

65. What is the minimum size schedule 80 PVC conduit required for all of the following conductors?

6 - #3 THWN

3 - #8 THW

2 - #10 THW

- (a) 2" (b) 2 1/2" (c) 1 1/2" (d) 3"

66. How many #8 THW conductors can you install in a 1" rigid steel conduit, 20" in length?

- (a) 12 (b) 9 (c) 6 (d) none of these

67. What size branch circuit inverse time breaker is required for a 3 hp, 230v, single-phase motor?

- (a) 20a (b) 25a (c) 30a (d) 45a

68. What is the maximum number of #6 XHHW conductors permitted in a 1 1/2" IMC conduit?

- (a) 11 (b) 13 (c) 15 (d) 14

69. In an apartment building, what is the maximum demand for 28 - 8 kW ranges?

- (a) 43 kW (b) 53.76 kW (c) 60 kW (d) none of these

70. The total load of a 15 hp, 208v, three-phase induction motor is closest to ____ va.

- (a) 10,000 (b) 11,000 (c) 15,000 (d) 17,000